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UNIT-3

CAPSULES

Capsule

It is a unit solid dosage form meant for oral use in which medicament are enclosed in a shell made up of gelatin eg. Vitamins, Amoxicillin etc.



Adv

- they are easy to manufacture because they avoid many unit operations such as compression, granulation & drying.
- Bioavailability is increases (↑ soft gelatin capsules).
- the taste of the bitter substance can be masked by encapsulating them in capsule form.
- they are elegant in appearance because available in wide range of colors.

Dis

- Not suitable for highly soluble substance like potassium chloride, potassium bromide etc.
- Not suitable for highly hygroscopic substances etc.

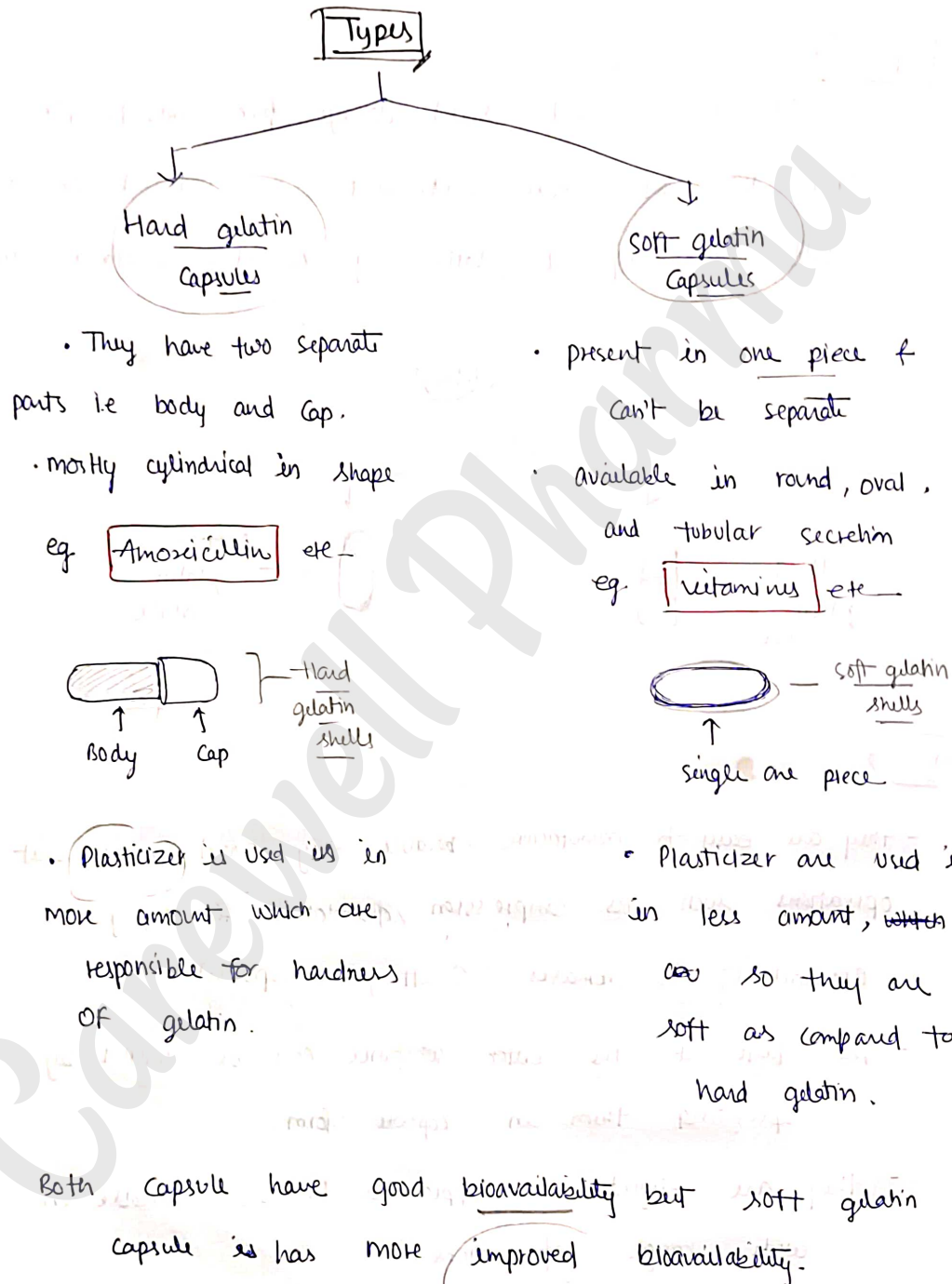
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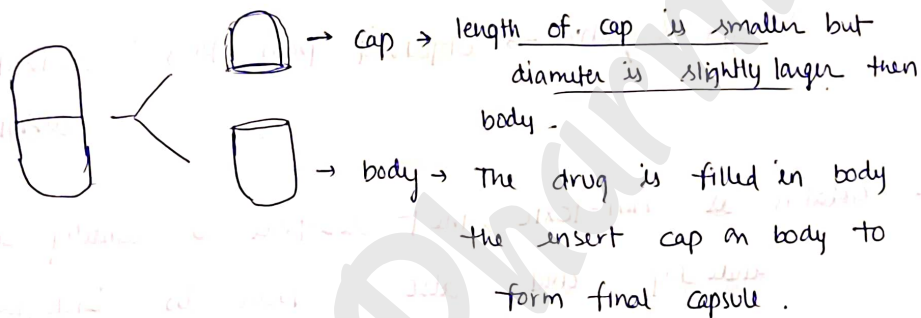
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Imp
(K)

write about production of hard gelatin capsule shells and also its process of filling

→ Hard gelatin capsule shells consists of two parts namely → "cap" and "body".



• formulation of gelatin shells →

Gelatin shells contain ingredients

- - Gelatin (main ingredients)
- - Plasticizers
- - Opacifying agents — to increase opacity
- - colouring agent
- - flavouring agent
- - Sweetening agents
- - Preservatives — parabens
- - Wetting agents

1) Gelatin → It is a tasteless, colorless protein substance which is extracted from from the tissue of animals.
 skin bone

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- Source

- Animals bones
 - Animals skin
- } Connective tissue → collagen

• Gelatin

- Type A gelatin → obtained from bones & derived from acid treatment.
- Type B gelatin → obtained from skins & derived from alkali (basic) treatment.

- Gelatin is non-toxic, inert substance so widely used in industry and also soluble in biological fluids and it also have good film forming properties.

ii) Plasticizer →

It increases the plasticity of the gelatin film. responsible for hardness.

eg. Sorbitol, glycerine, propylene glycol etc..

iii) Colouring agents →

provide color to the body & cap parts.

iv) Flavouring agents →

provide flavours to shell

v) Sweetening →

provide sweetness but not more than 5%.

vi) Opacifying agents →

Used to give opacity to the gelatin film. eg. Titanium dioxide reduces the transparency

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vii) preservatives →
prevent from microbes & increase life
eg. parabens etc.

viii) wetting agents →
used to wet the gelatin for make solⁿ.

- Manufacturing

- firstly, All raw material should be sterilized and released by quality test.
- A gelatin solution is prepared and transfer it into stainless steel feed tanks.
- Now, dyes, opacifiers, preservatives and water are added to the gelatin solⁿ in a feed tank. ^{plasticizers}

Now,

- Prepare stainless steel pins to make caps & bodies.
choose required sizes of these pins.
- Maintain high temperature (required)

Steps →

- 1) Dipping → Pins are lubricated and dip (enter) into gelatin filled container. (for both cap & bodies simultaneously).
Now, films are formed on the surface of each pin moulds.

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ii) Rotation / spinning →

Now, pins are removed from the container and during this they rotate until they are facing upward.

iii) Drying →

Now, they dry by the mechanism of hot air or oven.
excess water is removed during this process.

iv) Stripping →

Now, the capsule shells are removed from the pins with the help of bronze jaws (softer than stainless steel).

v) Trimming →

In this, excess length of capsule are removed with the help of stationary knives.

vi) Joining →

Now, both the half ^(halves) of the capsules are obtained and then they joined and ejected from the machine.

vii) Polishing →

Acela-cota pan are used to polish capsules and rubbed with clothes.

— used to increase the appearance

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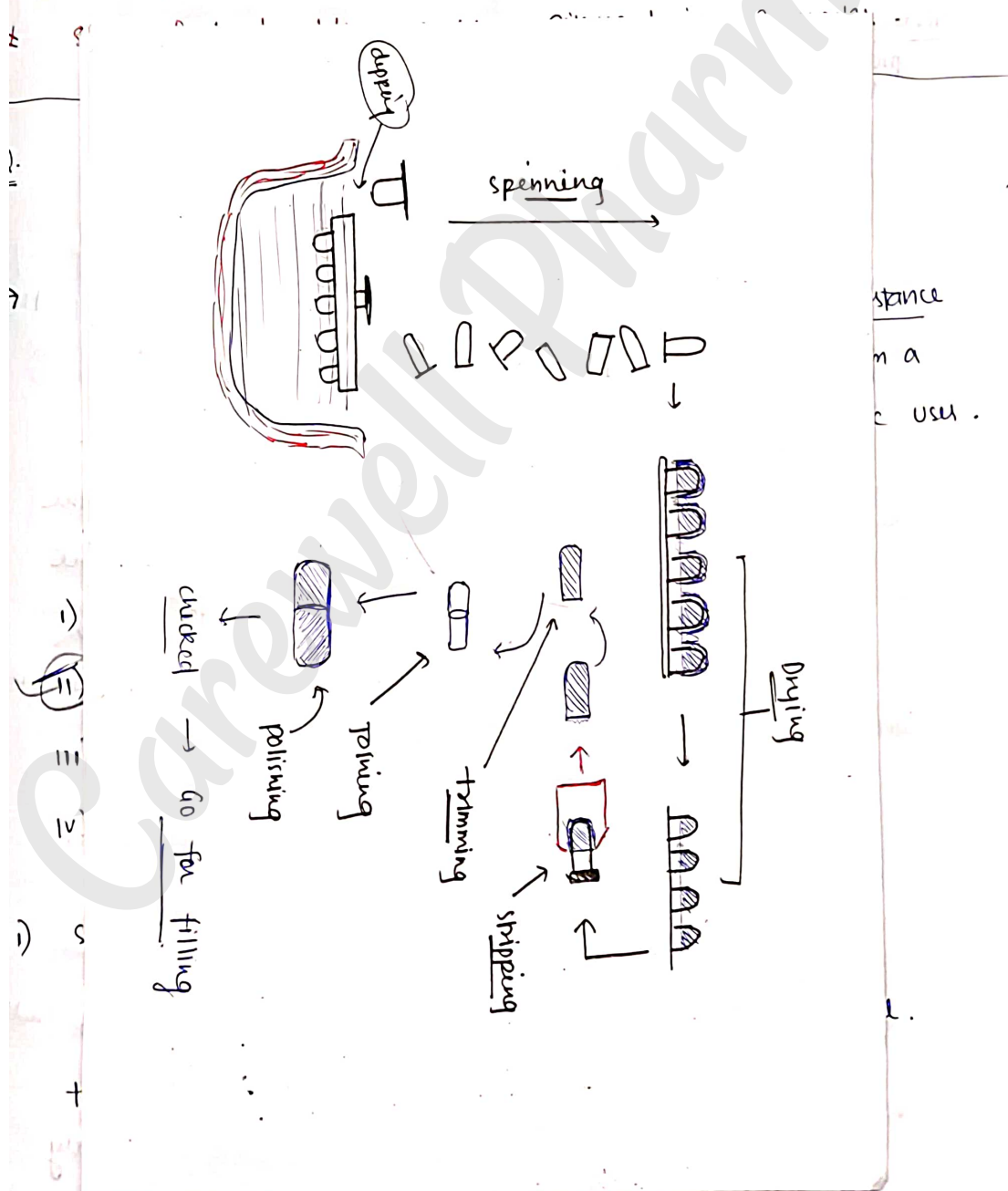
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lastly capsule are pushed into a conveyor belt.
 Capsule quality is monitored throughout the production
 process → capsules visually inspect and sorted.



Now, Ready to be sterilized and ~~for~~ packaged.



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sizes of hard gelatin capsules → Discussed in 2 marks.

Q. Write in details about process of filling hard gelatin capsule and also equipment used.

→ filling → It is a process in which a drug substance is filled in a gelatin shell and ~~not~~ form a final capsule which used for therapeutic use.

• Several steps for filling capsules

- i) selection of appropriate sizes
- ii) filling of hard-gelatin capsule
- iii) sealing of capsule
- iv) finishing of capsule

i) selection of appropriate sizes →

choose a required size of gelatin shell.

there are total eight sizes.

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ii) filling →

usually filled with drug in the form of
powder or granules.

Drug substance / content

(it includes API, fillers (dilvents), glidants etc.)

General

procedure →

- firstly prepare the capsule content (drug material).

↓

- Now load the empty gelatin shells in a way that cap is on top & body is on bottom.

↓

- Rectification → Now, after loading capsule check the all capsule shells that ~~are~~ they are fitted well and body end downward. Also check the defectiveness of capsule, size of capsule etc.

↓

- Now separate the cap from body and open the gelatin shell.

↓

Now, fill the body's part of capsule gelatin shell.

↓

Now, Again attach or join the cap on body then eject the capsule.

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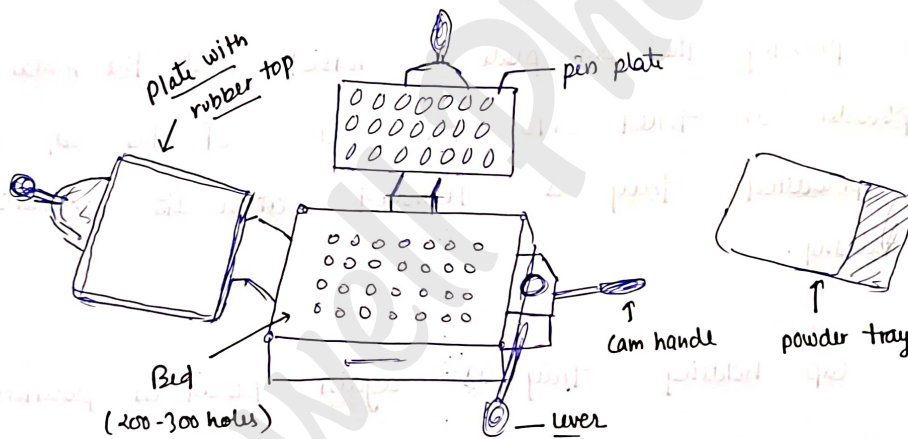
machines (equipment) →

Hand operated capsule filling machine

- Semi automatic capsule filling machine
- Automatic capsule filling machine

1) Hand operated capsule filling machine

— used for laboratory purpose



- It consists of a bed having 200-300 hole, a loading tray having 200-300 holes, a powder tray, a pin plate having 200-300 pins, a sealing plate having a rubber top, a lever, a cam handle

Procedure →

empty capsule are filled into loading tray and it is placed over the bed. The cam handle is operated to separate the capsule caps from their bodies



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The powder tray is placed in a proper position
and filled with an accurate quantity of powder
with scraper.



The excess of the powder is collected on the platform
of the powder tray, the pin plate is lowered and
the filled powder is pressed by moving the pin
downward



After pressing the pin plate is raised and the remaining
powder is filled into the bodies of the capsule.
The powdered tray is removed after its complete
filling.



The cap holding tray is again placed in position,
then the plate with the rubber top is lowered and
the lever is operated to lock the caps & bodies.



The loading tray is then removed and filled
capsule are collected.

-
- ii) Semi automatic capsule filling machine
→ Combined both automatic & manual methods
procedure → almost same

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• SOFT GELATIN CAPSULES

- ~~Almost same as hard but~~

Solid unit dosage form in which, liquid, semisolid or paste form of drug is filled inside the gelatin layer.

production →

four methods

1) Plate process →

- place the gelatin sheet over a die plate containing numerous die pockets

↓
application of vacuum to draw the sheet in to the die pockets.

↓
fill the pockets with liquid or paste

↓
Place another gelatin sheet over the filled pockets

↓
Sandwich under a die press where the capsules are formed and cut out.

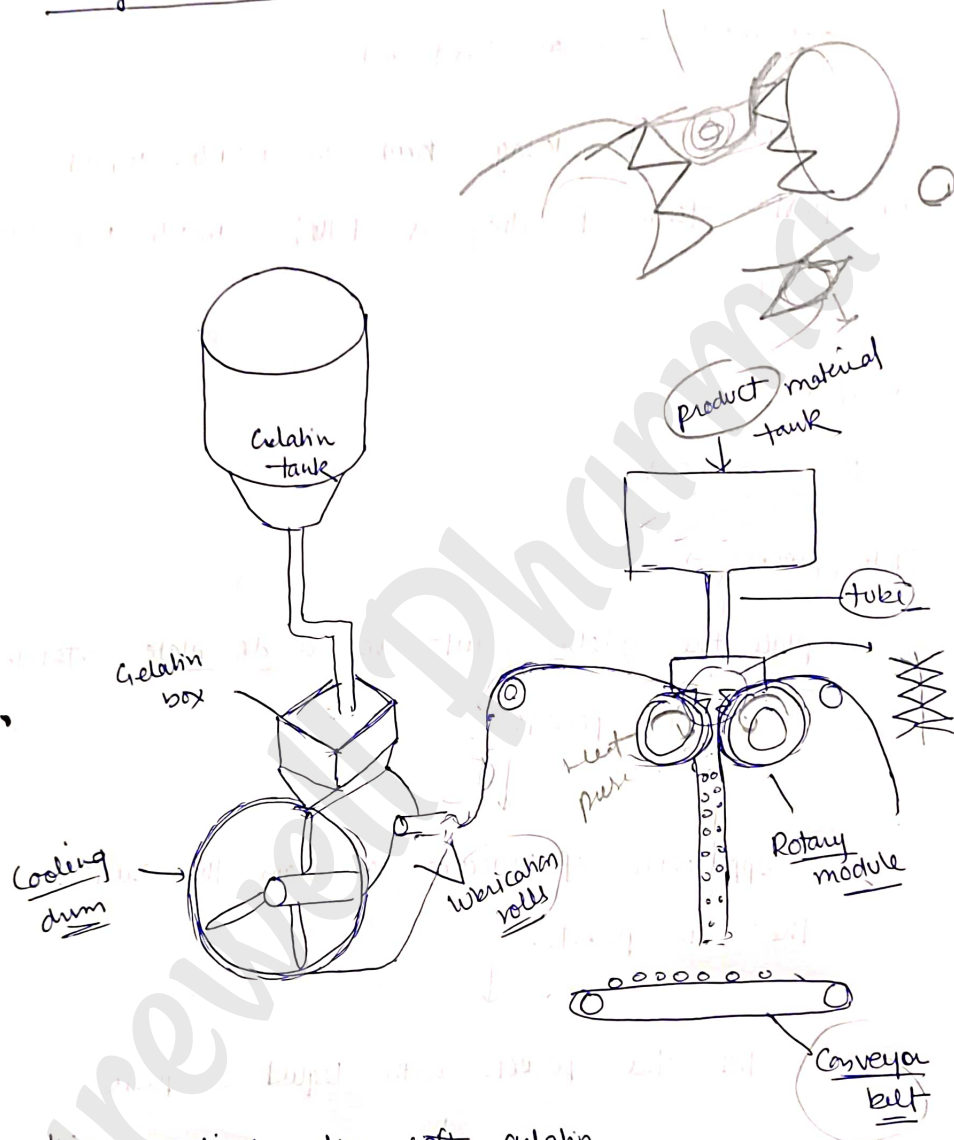
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2) Rotary die process →



- In this machine the soft gelatin capsule are prepared and then immediately with liquid medicament, it is having two hopper and two rotating dies.
- liquid mixture is placed in one hopper and liquid medicament in other
- the two rotating dies rotates in opposite direction when the fluid gelatin mixture enter

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the machine from the hopper it produce two continuous ribbon.

- these half shells of the capsule are formed.
- At this stage the measured quantity of the medicament is filled into it with the stroke of a pump with the subsequent movement of the dies, the other half capsule is formed.
- The two halves of the capsules are sealed together by the heat and pressure of the rotating dies.
- As the die rolls rotates, the convergence of the matching die pockets seals and cuts out the filled capsules.

3) Accogel machine /stern machine

↳ used to fill dry powder into soft gelatin Capsule

• Quality control tests →

- Disintegration test
- Dissolution test
- weight variation —
- Content uniformity —

• Bloom strength of gelatin → only for hard gelatin

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